

K -theory of the Roe algebra

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This is the working note of an ongoing project and also expository notes from *Higher Index Theory*.
We try to keep the note somewhat self-contained.

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1 Basic definitions

We work with proper metric spaces when discussing Roe algebras, and with locally compact, second countable, Hausdorff spaces when discussing geometric modules and localisation algebras.

Roe and localised Roe algebras are defined using geometric modules; after passing to K -theory, the constructions are independent of the ample module up to the canonical identifications used in [WY20, Chs. 5–7].

1.1 Geometric modules, support, and propagation

Definition 1.1 ([WY20, Def. 4.1.1]). Let X be a locally compact, second countable, Hausdorff space. A *geometric X module* is a separable Hilbert space H_X equipped with a non-degenerate representation $\rho : C_0(X) \longrightarrow B(H_X)$. It is *ample* if no non-zero element of $C_0(X)$ acts as a compact operator, and if H_X is infinite-dimensional.

Definition 1.2 ([WY20, Defs. 4.1.7 and 4.1.8]). Let H_X and H_Y be geometric modules over locally compact spaces X and Y , and let $T : H_X \rightarrow H_Y$ be bounded. The *support* of T , denoted $\text{supp}(T) \subseteq Y \times X$, is the set of all pairs (y, x) such that

$$\chi_V T \chi_U \neq 0$$

for every pair of open neighbourhoods $V \ni y$ and $U \ni x$. If $X = Y$ is metric and $T \in B(H_X)$, the *propagation* of T is

$$\text{prop}(T) := \sup\{d(y, x) : (y, x) \in \text{supp}(T)\} \in [0, \infty].$$

Definition 1.3 ([WY20, Def. 5.1.1]). Let H_X be a geometric module over a metric space X . A bounded operator $T \in B(H_X)$ is *locally compact* if, for every compact subset $K \subseteq X$,

$$\chi_K T \quad \text{and} \quad T \chi_K$$

are compact. It has *finite propagation* if $\text{prop}(T) < \infty$.

Definition 1.4 ([WY20, Def. 5.1.4]). The *Roe $*$ -algebra* of H_X is

$$C[H_X] := \{T \in B(H_X) : T \text{ is locally compact and has finite propagation}\}.$$

The *Roe algebra* of H_X is the norm closure of $C[H_X]$ in $B(H_X)$, and is denoted $C^*(H_X)$. If we have fixed an ample module over X , we write simply $C^*(X)$.

We use the definitions of $K_0(A)$ and $K_1(A)$ from [WY20, Ch. 2].

1.2 Localisation and localised Roe algebras

Definition 1.5 ([WY20, Defs. 6.2.1 and 6.2.3]). Let H_X be an X module, and let X^+ denote the one-point compactification of X . Define $L[H_X]$ to be the $*$ -algebra of all bounded functions

$$(T_t)_{t \in [1, \infty)} : [1, \infty) \longrightarrow B(H_X)$$

such that:

(i) for every compact $K \subseteq X$, there is $t_K \geq 1$ such that, for all $t \geq t_K$,

$$\chi_K T_t \quad \text{and} \quad T_t \chi_K$$

are compact, and the functions

$$t \mapsto \chi_K T_t, \quad t \mapsto T_t \chi_K$$

are uniformly norm continuous on $[t_K, \infty)$;

(ii) for every open neighbourhood U of the diagonal in $X^+ \times X^+$, there is $t_U \geq 1$ such that, for all $t > t_U$,

$$\text{supp}(T_t) \subseteq U.$$

The *localisation algebra* $L^*(H_X)$ is the C^* -algebra completion of $L[H_X]$ for the supremum norm

$$\|(T_t)\| := \sup_{t \in [1, \infty)} \|T_t\|_{B(H_X)}.$$

If H_X is ample, we write $L^*(X) := L^*(H_X)$.

Definition 1.6 ([WY20, Def. 6.3.1]). If H_X is ample, the K -homology groups of X are

$$K_n(X) := K_n(L^*(X)), \quad n \geq 0.$$

Thus $K_*(X) = K_0(X) \oplus K_1(X)$.

Definition 1.7 ([WY20, Def. 6.6.1]). Let H_X be a geometric module over a proper metric space X . The *localised Roe $*$ -algebra* $C_L[H_X]$ is the $*$ -algebra of all uniformly continuous, bounded functions

$$(T_t)_{t \in [1, \infty)} : [1, \infty) \longrightarrow C[H_X]$$

such that, for every open neighbourhood U of the diagonal in $X^+ \times X^+$, there is $t_U \geq 1$ such that, for all $t > t_U$,

$$\text{supp}(T_t) \subseteq U.$$

The *localised Roe algebra* $C_L^*(H_X)$ is the completion of $C_L[H_X]$ in the supremum norm.

If we fix an ample module over X , we abbreviate $C^*(H_X)$ and $C_L^*(H_X)$ to $C^*(X)$ and $C_L^*(X)$.

Proposition 1.8 ([WY20, Prop. 6.6.2]). If H_X is an ample geometric module, then the natural inclusion

$$C_L^*(H_X) \longrightarrow L^*(H_X)$$

induces an isomorphism on K -theory. In particular,

$$K_*(C_L^*(H_X)) \cong K_*(X),$$

where the right-hand side is the analytic K -homology of X .

Lemma 1.9 ([WY20, Lem. 5.1.8]). If X is bounded, then

$$C^*(H_X) = \mathcal{K}(H_X)$$

for every X module H_X .

2 Relative Roe algebras and full-corner identifications

2.1 The relative algebras

Let X be a proper metric space and let $E \subseteq X$ be closed. Choose a countable dense set $Z \subseteq X$ such that $Z \cap E$ is dense in E .

When a cover $X = E \cup F$ is used below, choose countable dense subsets $Z_E \subseteq E$ and $Z_F \subseteq F$ with $Z_E \cap Z_F$ dense in $E \cap F$, and put $Z := Z_E \cup Z_F$, as in the proof of [WY20, Thm. 7.5.5].

Let H be a separable infinite-dimensional Hilbert space. Put $H_X := \ell^2(Z, H)$ and $H_E := \ell^2(Z \cap E, H) = \chi_E H_X$. We write χ_E both for the characteristic function of E and for the corresponding projection $H_X \rightarrow H_E$.

Definition 2.1 (cf. [WY20, proof of Thm. 7.5.5]). Define the C^* -subalgebra

$$C_X^*(E) := \overline{\text{span}\{S\chi_E T : S, T \in C^*(X)\}} \subseteq C^*(X).$$

Equivalently, $C_X^*(E)$ is the ideal of $C^*(X)$ generated by the multiplier χ_E . Similarly,

$$C_{L,X}^*(E) := \overline{\text{span}\{(S_t)\chi_E(T_t) : (S_t), (T_t) \in C_L^*(X)\}} \subseteq C_L^*(X)$$

is the ideal generated by the multiplier χ_E in the localised Roe algebra.

The algebra $C_X^*(E)$ can also be described as the closure of the finite-propagation, locally compact operators on H_X supported in some neighbourhood of E , i.e. operators T for which

$$\chi_{N_r(E)} T = T \chi_{N_r(E)} = T$$

for some $r > 0$, where $N_r(E) := \{x \in X : d(x, E) \leq r\}$. We have analogous constructions for $F \subseteq X$ and $E \cap F \subseteq X$.

2.2 Extension by zero and full corners

Write $H_X = H_E \oplus H_E^\perp$. If $T \in C^*(E)$, its *extension by zero* to H_X is the operator

$$\tilde{T} = \begin{pmatrix} T & 0 \\ 0 & 0 \end{pmatrix}.$$

Finite propagation for the subspace metric on E remains finite propagation in X , and local compactness is preserved under extension by zero.

Proposition 2.2 (cf. [WY20, Prop. 2.7.19 and proof of Thm. 7.5.5]). *The extension-by-zero map identifies $C^*(E)$ with the corner*

$$\chi_E C_X^*(E) \chi_E.$$

Moreover, this corner is full in $C_X^(E)$. Consequently, the inclusion*

$$C^*(E) \cong \chi_E C_X^*(E) \chi_E \hookrightarrow C_X^*(E)$$

induces an isomorphism on K -theory.

Proof. If $T \in C^*(E)$, then $\tilde{T} \in C^*(X)$ and

$$\tilde{T} = \chi_E \tilde{T} \chi_E,$$

so the image lies in $\chi_E C_X^*(E) \chi_E$.

Conversely, let $a \in \chi_E C_X^*(E) \chi_E$. Since a vanishes on $H_E^\perp$ and preserves H_E , there exists a bounded operator $T : H_E \rightarrow H_E$ with

$$a = \begin{pmatrix} T & 0 \\ 0 & 0 \end{pmatrix}.$$

On the dense $*$ -subalgebra generated by finite-propagation terms of the form $S \chi_E U$, compression to H_E produces finite-propagation, locally compact operators on H_E . Passing to the norm closure gives $T \in C^*(E)$.

For fullness, a dense generating set for $C_X^*(E)$ is spanned by terms $S \chi_E T$. But

$$S \chi_E T = (S \chi_E) \chi_E (\chi_E T) \in C_X^*(E) \chi_E C_X^*(E).$$

Thus the ideal generated by the corner projection χ_E contains a dense generating set for $C_X^*(E)$. The K -theory statement follows because inclusion of a full corner induces an isomorphism on K -theory [WY20, Prop. 2.7.19]. $\square$

Proposition 2.3 (cf. [WY20, Prop. 2.7.19 and proof of Thm. 7.5.5]). *After fixing the extension-by-zero embedding, $C_L^*(E)$ identifies with the full corner*

$$\chi_E C_{L,X}^*(E) \chi_E \subseteq C_{L,X}^*(E).$$

Consequently, the inclusion

$$C_L^*(E) \cong \chi_E C_{L,X}^*(E) \chi_E \hookrightarrow C_{L,X}^*(E)$$

induces an isomorphism

$$K_*(C_L^*(E)) \cong K_*(C_{L,X}^*(E)).$$

The same statements hold with E replaced by F or $E \cap F$.

Proof. Apply the previous argument first to the dense $*$ -algebras and then pass to the supremum-norm completions. If $(T_t) \in C_L[H_E]$, then extension by zero gives a family

$$\tilde{T}_t = \begin{pmatrix} T_t & 0 \\ 0 & 0 \end{pmatrix}$$

that is uniformly continuous, locally compact, and has support shrinking to the diagonal in $X^+ \times X^+$.

Conversely, compression to H_E recovers a family in $C_L^*(E)$, since neighbourhoods of the diagonal in $E^+ \times E^+$ are obtained by restricting neighbourhoods of the diagonal in $X^+ \times X^+$. For fullness, each generator $(S_t) \chi_E (T_t)$ belongs to $C_{L,X}^*(E) \chi_E C_{L,X}^*(E)$.

The K -theory statement follows from [WY20, Prop. 2.7.19]. $\square$

3 Coarsely excisive decompositions and Mayer–Vietoris

3.1 Coarsely excisive covers

Definition 3.1 ([WY20, Def. 7.5.3]). Let $X = E \cup F$ be a cover by closed subsets. The cover is *coarsely excisive* if, for every $r > 0$, there exists $s > 0$ such that

$$N_r(E) \cap N_r(F) \subseteq N_s(E \cap F).$$

Example 3.2 ([WY20, Example 7.5.4]). For

$$E = \mathbb{R}^{n-1} \times (-\infty, 0], \quad F = \mathbb{R}^{n-1} \times [0, \infty),$$

we have $E \cap F = \mathbb{R}^{n-1} \times \{0\}$, and

$$N_r(E) \cap N_r(F) = \mathbb{R}^{n-1} \times [-r, r] \subseteq N_r(E \cap F).$$

So this cover is coarsely excisive.

3.2 Intersection and sum identities

Proposition 3.3 (cf. [WY20, proof of Thm. 7.5.5]). *If $X = E \cup F$ is coarsely excisive, then*

$$C_X^*(E) \cap C_X^*(F) = C_X^*(E \cap F), \quad C_X^*(E) + C_X^*(F) = C^*(X),$$

and similarly

$$C_{L,X}^*(E) \cap C_{L,X}^*(F) = C_{L,X}^*(E \cap F), \quad C_{L,X}^*(E) + C_{L,X}^*(F) = C_L^*(X).$$

Proof. The inclusion $C_X^*(E \cap F) \subseteq C_X^*(E) \cap C_X^*(F)$ is immediate.

For the reverse inclusion, recall that the intersection of two closed C^* -ideals is the closed linear span of their products. A product of finite-propagation operators supported near E and F , respectively, is supported in the intersection of suitably enlarged neighbourhoods of E and F . Coarse excision places this intersection in a neighbourhood of $E \cap F$. Approximating the two factors in their relative dense subalgebras and taking closures gives the reverse inclusion.

For the sum identity, use the multiplier decomposition $T = \chi_E T + (1 - \chi_E)T$. The first term lies in $C_X^*(E)$, and since $X = E \cup F$, the second term is supported over F and lies in $C_X^*(F)$. Hence $C_X^*(E) + C_X^*(F) = C^*(X)$. For the localised intersection identity, use the closedness argument in the proof of [WY20, Thm. 7.5.5]; the same multiplier decomposition gives the localised sum identity. $\square$

3.3 The pushout cube and the six-term sequences

Because of proposition 3.3, the relative algebras fit into a commutative cube of pushout diagrams. The left- and right-hand vertices record both the original intersection/sum expressions and the simplifications from proposition 3.3.

$$\begin{array}{ccccc}
C_{L,X}^*(E) \cap C_{L,X}^*(F) & \xrightarrow{\quad} & C_{L,X}^*(E) & \xrightarrow{\quad} & C_X^*(E) \\
= C_{L,X}^*(E \cap F) & \searrow \text{ev}_\nu & & & \\
& & C_X^*(E) \cap C_X^*(F) & \xrightarrow{\quad} & \\
& & = C_X^*(E \cap F) & & \\
& & \downarrow & & \\
C_{L,X}^*(F) & \xrightarrow{\quad} & C_{L,X}^*(E) + C_{L,X}^*(F) & \xrightarrow{\quad} & C_X^*(E) + C_X^*(F) \\
& \searrow \text{ev}_\nu & = C_L^*(X) & & \\
& & \downarrow & & \\
& & C_X^*(F) & \xrightarrow{\quad} & C_X^*(E) + C_X^*(F) \\
& & & & = C^*(X)
\end{array}$$

We apply [WY20, Prop. 2.7.15] once to the top pushout square and once to the bottom pushout square. The evaluation-at-one maps assemble these into a natural comparison between the localised and non-localised Mayer–Vietoris sequences. Then the full-corner identifications in propositions 2.2 and 2.3 replace, at the level of K -theory, the relative algebras by the Roe and localised Roe algebras of E , F , and $E \cap F$.

Theorem 3.4 ([WY20, Thm. 7.5.5]). *Let $X = E \cup F$ be a coarsely excisive decomposition of a proper metric space. Then there are six-term exact sequences*

$$\begin{array}{ccccc}
K_0(C^*(E \cap F)) & \longrightarrow & K_0(C^*(E)) \oplus K_0(C^*(F)) & \longrightarrow & K_0(C^*(X)) \\
\uparrow \partial_1 & & & & \downarrow \partial_0 \\
K_1(C^*(X)) & \longleftarrow & K_1(C^*(E)) \oplus K_1(C^*(F)) & \longleftarrow & K_1(C^*(E \cap F))
\end{array}$$

and

$$\begin{array}{ccccc}
K_0(C_L^*(E \cap F)) & \longrightarrow & K_0(C_L^*(E)) \oplus K_0(C_L^*(F)) & \longrightarrow & K_0(C_L^*(X)) \\
\uparrow \partial_1^L & & & & \downarrow \partial_0^L \\
K_1(C_L^*(X)) & \longleftarrow & K_1(C_L^*(E)) \oplus K_1(C_L^*(F)) & \longleftarrow & K_1(C_L^*(E \cap F)).
\end{array}$$

Moreover, for each $i \in \{0, 1\}$, the evaluation-at-one map gives a commutative diagram

$$\begin{array}{ccccc}
K_i(C_L^*(E \cap F)) & \longrightarrow & K_i(C_L^*(E)) \oplus K_i(C_L^*(F)) & \longrightarrow & K_i(C_L^*(X)) \\
\downarrow \text{ev}_* & & \downarrow \text{ev}_* \oplus \text{ev}_* & & \downarrow \text{ev}_* \\
K_i(C^*(E \cap F)) & \longrightarrow & K_i(C^*(E)) \oplus K_i(C^*(F)) & \longrightarrow & K_i(C^*(X)).
\end{array}$$

Proof. Apply [WY20, Prop. 2.7.15] to the top pushout square of the cube and again to the bottom pushout square. This gives a commutative diagram of Mayer–Vietoris sequences for $C_{L,X}^*(E)$, $C_{L,X}^*(F)$, $C_{L,X}^*(E \cap F)$ and for $C_X^*(E)$, $C_X^*(F)$, $C_X^*(E \cap F)$.

The corner identifications in propositions 2.2 and 2.3 replace, at the level of K -theory, the relative algebras $C_X^*(E)$, $C_X^*(F)$, $C_X^*(E \cap F)$ and $C_{L,X}^*(E)$, $C_{L,X}^*(F)$, $C_{L,X}^*(E \cap F)$ by the Roe and localised Roe algebras of E , F , and $E \cap F$. After these identifications, the relative Mayer–Vietoris sequences become the two six-term sequences above, and naturality gives the commutative diagram. $\square$

4 The half-space Eilenberg swindle

Proposition 4.1 ([WY20, Prop. 7.5.2]). *Let $Y := X \times [0, \infty)$ with the product metric*

$$d((x, t), (x', t')) = \sqrt{d_X(x, x')^2 + |t - t'|^2}.$$

Then

$$K_*(C^*(Y)) = 0.$$

Proof. Choose an ample module of the form $H_Y = H_X \otimes L^2([0, \infty))$. For each $n \geq 1$, let $V_n : L^2([0, \infty)) \rightarrow L^2([0, \infty))$ be the right-shift isometry

$$(V_n u)(t) = \begin{cases} u(t - n), & t \geq n, \\ 0, & 0 \leq t < n. \end{cases}$$

Define

$$H_Y^\infty := \bigoplus_{n \geq 1} H_Y,$$

which is again an ample Y module, and let $W_n : H_Y \rightarrow H_Y^\infty$ denote the inclusion into the n -th summand. First define Φ on the dense Roe $*$ -algebra $C[H_Y]$. For $T \in C[H_Y]$, put

$$\Phi(T) := \sum_{n \geq 1} W_n (1 \otimes V_n) T (1 \otimes V_n^*) W_n^*.$$

Each summand is a translate of T along the half-line. Hence a compact subset of Y meets only finitely many of these translates, so $\Phi(T)$ is locally compact. Since the ranges of the W_n are orthogonal, the sum defines a bounded operator and $\|\Phi(T)\| \leq \|T\|$. Finite propagation is preserved under conjugation by $1 \otimes V_n$, so $\Phi(T) \in C[H_Y^\infty]$. Thus Φ is a contractive $*$ -homomorphism on $C[H_Y]$, and it extends by continuity to

$$\Phi : C^*(Y) \longrightarrow C^*(H_Y^\infty).$$

Let $\Psi : C^*(Y) \rightarrow C^*(H_Y^\infty)$ be the map that places an operator into the first summand, $\Psi(T) = W_1 T W_1^*$. For every $T \in C^*(Y)$, the elements

$$\begin{pmatrix} \Phi(T) & 0 \\ 0 & \Psi(T) \end{pmatrix}, \quad \begin{pmatrix} \Phi(T) & 0 \\ 0 & 0 \end{pmatrix}$$

of $M_2(C^*(H_Y^\infty))$ are conjugate by a fixed isometry in the multiplier algebra of $M_2(C^*(H_Y^\infty))$. By [WY20, Prop. 2.7.5],

$$\Phi_* + \Psi_* = \Phi_*$$

as maps $K_*(C^*(Y)) \longrightarrow K_*(C^*(H_Y^\infty))$. Thus $\Psi_* = 0$.

On the other hand, Ψ is induced by a covering isometry for the identity coarse map $Y \rightarrow Y$, so Ψ_* is an isomorphism by [WY20, Thm. 5.1.15]. Hence $K_*(C^*(Y)) = 0$. $\square$

Corollary 4.2 ([WY20, Prop. 7.5.2]). *For the same $Y = X \times [0, \infty)$,*

$$K_*(C_L^*(Y)) = 0.$$

Consequently, the assembly map for Y is an isomorphism.

Proof. By [WY20, Prop. 6.4.14], the K -homology of $X \times [0, \infty)$ is zero. By proposition 1.8, this is canonically isomorphic to $K_*(C_L^*(Y))$. The assembly map is therefore an isomorphism because both its domain and codomain vanish. $\square$

5 The Roe algebra K -theory of Euclidean space

5.1 The inductive step

For $n \geq 1$, let

$$E_n := \mathbb{R}^{n-1} \times (-\infty, 0], \quad F_n := \mathbb{R}^{n-1} \times [0, \infty).$$

Then

$$\mathbb{R}^n = E_n \cup F_n, \quad E_n \cap F_n \cong \mathbb{R}^{n-1},$$

and example 3.2 shows that this cover is coarsely excisive. Moreover,

$$E_n \cong \mathbb{R}^{n-1} \times [0, \infty), \quad F_n \cong \mathbb{R}^{n-1} \times [0, \infty),$$

so proposition 4.1 and corollary 4.2 give

$$K_*(C^*(E_n)) = K_*(C^*(F_n)) = 0, \quad K_*(C_L^*(E_n)) = K_*(C_L^*(F_n)) = 0.$$

Proposition 5.1 (cf. [WY20, Thm. 7.5.5]). *For every $n \geq 1$, the Mayer–Vietoris boundary maps give isomorphisms*

$$K_0(C^*(\mathbb{R}^n)) \cong K_1(C^*(\mathbb{R}^{n-1})), \quad K_1(C^*(\mathbb{R}^n)) \cong K_0(C^*(\mathbb{R}^{n-1})).$$

The same statement holds with C^ replaced by C_L^* .*

Proof. Apply theorem 3.4 to $\mathbb{R}^n = E_n \cup F_n$. Since the K -theory of their Roe algebras vanishes, the six-term exact sequence collapses to

$$\begin{array}{ccccc} K_0(C^*(\mathbb{R}^{n-1})) & \longrightarrow & 0 & \longrightarrow & K_0(C^*(\mathbb{R}^n)) \\ \uparrow & & & & \downarrow \\ K_1(C^*(\mathbb{R}^n)) & \longleftarrow & 0 & \longleftarrow & K_1(C^*(\mathbb{R}^{n-1})), \end{array}$$

from which the isomorphisms are immediate. The localised case is identical. $\square$

Example 5.2. Proposition 5.1 already shows that

$$K_0(C^*(\mathbb{R})) = 0, \quad K_1(C^*(\mathbb{R})) \cong \mathbb{Z},$$

and then

$$K_0(C^*(\mathbb{R}^2)) \cong \mathbb{Z}, \quad K_1(C^*(\mathbb{R}^2)) = 0.$$

5.2 The computation

Theorem 5.3 (cf. [WY20, proof of Thm. 7.5.1]). *For every $n \geq 0$ and $j \in \{0, 1\}$,*

$$K_j(C^*(\mathbb{R}^n)) \cong \begin{cases} \mathbb{Z}, & j \equiv n \pmod{2}, \\ 0, & j \not\equiv n \pmod{2}. \end{cases}$$

Equivalently,

$$K_0(C^*(\mathbb{R}^n)) \cong \begin{cases} \mathbb{Z}, & n \text{ even}, \\ 0, & n \text{ odd}, \end{cases} \quad K_1(C^*(\mathbb{R}^n)) \cong \begin{cases} 0, & n \text{ even}, \\ \mathbb{Z}, & n \text{ odd}. \end{cases}$$

Proof. We argue by induction on n . If $n = 0$, then $\mathbb{R}^0$ is a point. By lemma 1.9, $C^*(\mathbb{R}^0) = \mathcal{K}$, and by [WY20, Example 2.7.3],

$$K_0(\mathcal{K}) \cong \mathbb{Z}, \quad K_1(\mathcal{K}) = 0.$$

This is the base case. The induction step follows from proposition 5.1. □

Corollary 5.4 ([WY20, Thm. 7.5.1]). *For every $n \geq 0$, the evaluation-at-one map*

$$\text{ev}_* : K_*(C_L^*(\mathbb{R}^n)) \longrightarrow K_*(C^*(\mathbb{R}^n))$$

is an isomorphism.

Proof. The case $n = 0$ is the assembly map for a point, hence an isomorphism. For the inductive step, apply theorem 3.4 to the cover $\mathbb{R}^n = E_n \cup F_n$. The terms for E_n and F_n vanish by proposition 4.1 and corollary 4.2, so the five lemma shows that if ev_* is an isomorphism for $\mathbb{R}^{n-1}$, then it is also an isomorphism for $\mathbb{R}^n$. □

Definition 5.5 (cf. [WY20, Thm. 7.5.5]). Let $p \in \mathcal{K}$ be a rank-one projection, and set

$$\beta_0 := [p] \in K_0(C^*(\mathbb{R}^0)) = K_0(\mathcal{K}).$$

For $n \geq 1$, define β_n recursively by

$$\partial_n(\beta_n) = \beta_{n-1},$$

where ∂_n is the Mayer–Vietoris boundary isomorphism for

$$\mathbb{R}^n = \mathbb{R}^{n-1} \times (-\infty, 0] \cup \mathbb{R}^{n-1} \times [0, \infty).$$

The sign convention is the one we use below: the positive half-space is the I -term in the pushout boundary. Reversing the order of the two half-spaces reverses the sign. Thus $\beta_n \in K_n(C^*(\mathbb{R}^n))$, where $K_n(A)$ is identified with $K_{n \bmod 2}(A)$ by Bott periodicity [WY20, Thm. 2.6.5].

Thus

$$\beta_n \xrightarrow{\partial_n} \beta_{n-1} \xrightarrow{\partial_{n-1}} \cdots \xrightarrow{\partial_1} \beta_0 = [p].$$

The rest of these notes identify this generator more concretely.

6 The boundary map for $\mathbb{R}$ and the one-dimensional shift

6.1 How the boundary map is built

Suppose

$$\begin{array}{ccc} I \cap J & \xhookrightarrow{\iota_I} & I \\ \downarrow \iota_J & & \downarrow \\ J & \xhookrightarrow{\quad} & A \end{array}$$

is a pushout diagram of ideals with $A = I + J$. Then the inclusion $I \hookrightarrow A$ induces an isomorphism

$$I/(I \cap J) \cong A/J.$$

Hence there are short exact sequences

$$0 \longrightarrow I \cap J \longrightarrow I \longrightarrow I/(I \cap J) \longrightarrow 0$$

and

$$0 \longrightarrow J \longrightarrow A \longrightarrow A/J \longrightarrow 0.$$

The odd Mayer–Vietoris boundary is the composite

$$K_1(A) \longrightarrow K_1(A/J) \cong K_1(I/(I \cap J)) \xrightarrow{\partial} K_0(I \cap J),$$

where the last arrow is the index map for the first short exact sequence. The even boundary is defined similarly, with K_0 and K_1 interchanged and the index map replaced by the exponential map of [WY20, Rem. 2.6.8]. These are the boundary maps in the six-term Mayer–Vietoris sequence of [WY20, Prop. 2.7.15].

6.2 The one-dimensional computation

Take

$$X = \mathbb{R}, \quad E = (-\infty, 0], \quad F = [0, \infty).$$

Then $E \cap F = \{0\}$, and the cover is coarsely excisive. By proposition 4.1,

$$K_*(C^*(E)) = K_*(C^*(F)) = 0.$$

Since $E \cap F$ is bounded, lemma 1.9 gives

$$C^*(E \cap F) = C^*(\{0\}) \cong \mathcal{K}.$$

Thus

$$K_0(C^*(E \cap F)) \cong \mathbb{Z}, \quad K_1(C^*(E \cap F)) = 0.$$

The six-term sequence simplifies to

$$\begin{array}{ccccc} K_0(\mathcal{K}) & \longrightarrow & 0 & \longrightarrow & K_0(C^*(\mathbb{R})) \\ \uparrow \partial_1 & & & & \downarrow \partial_0 \\ K_1(C^*(\mathbb{R})) & \longleftarrow & 0 & \longleftarrow & K_1(\mathcal{K}) = 0, \end{array}$$

so

$$K_0(C^*(\mathbb{R})) = 0, \quad \partial_1 : K_1(C^*(\mathbb{R})) \xrightarrow{\cong} K_0(\mathcal{K})$$

is an isomorphism, and ∂_0 is the zero map.

Proposition 6.1 ([WY20, Example 2.7.3 and Thm. 7.5.5]). *Let $p \in \mathcal{K}$ be a rank-one projection. Then $[p]$ generates $K_0(\mathcal{K}) \cong \mathbb{Z}$. If $\beta_1 \in K_1(C^*(\mathbb{R}))$ is the unique class satisfying*

$$\partial_1(\beta_1) = [p],$$

then β_1 is a generator of $K_1(C^(\mathbb{R})) \cong \mathbb{Z}$.*

6.3 Toeplitz compression and the sign

We discretize $\mathbb{R}$ by $\mathbb{Z}$ to understand the generator. Let $S : \ell^2(\mathbb{Z}) \rightarrow \ell^2(\mathbb{Z})$ be the bilateral shift

$$S\delta_n = \delta_{n+1}.$$

On the non-ampl module $\ell^2(\mathbb{Z})$, S is locally compact and has propagation one. On the ampl module $\ell^2(\mathbb{Z}, H)$, where H is separable infinite-dimensional, the shift $S \otimes 1_H$ is not locally compact, because the matrix entries are 1_H . Therefore we use a rank-one projection $e \in \mathcal{K}(H)$ and the unitary representatives

$$U_{S^{-1}} := 1 + (S^{-1} - 1) \otimes e, \quad U_S := 1 + (S - 1) \otimes e$$

in $C^*(\mathbb{Z})^+$. We write their K_1 -classes as $[S^{-1}]$ and $[S]$. Through the coarse equivalence $\mathbb{Z} \hookrightarrow \mathbb{R}$, which induces an isomorphism on Roe algebra K -theory by [WY20, Thm. 5.1.15], we regard these as classes in $K_1(C^*(\mathbb{R}))$.

Let $H_+ := \ell^2(\mathbb{N})$, where $\mathbb{N} = \{0, 1, 2, \dots\}$, and let

$$V : H_+ \hookrightarrow \ell^2(\mathbb{Z}), \quad V\delta_n = \delta_n.$$

From the Toeplitz discussion in [WY20, Sec. 3.4], if $f \in C(S^1)$ and C_f is the corresponding convolution operator, then

$$T_f := V^* C_f V$$

is the Toeplitz operator on H_+ .

For $f(z) = z$, the convolution operator is S , so

$$T_z = V^* S V.$$

Thus T_z is the forward unilateral shift:

$$T_z \delta_n = \delta_{n+1}.$$

For $f(z) = z^{-1}$, the convolution operator is S^{-1} , so

$$T_{z^{-1}} = V^* S^{-1} V = T_z^*.$$

Thus T_z^* is the backward unilateral shift:

$$T_z^* \delta_0 = 0, \quad T_z^* \delta_n = \delta_{n-1} \quad (n \geq 1).$$

Let p be the rank-one projection onto $\mathbb{C}\delta_0 \subset H_+$. Then

$$T_z^* T_z = 1, \quad T_z T_z^* = 1 - p.$$

Therefore

$$\ker(T_z) = 0, \quad \operatorname{coker}(T_z) \cong \mathbb{C}\delta_0,$$

and

$$\operatorname{Index}(T_z) = \dim \ker(T_z) - \dim \operatorname{coker}(T_z) = -1.$$

In $K_0(\mathcal{K})$ this says that the K -theory index class of T_z is $-[p]$. This agrees with the Toeplitz index theorem

$$\operatorname{Index}(T_f) = -\operatorname{wind}(f)$$

from [WY20, Thm. 3.4.1], valid for nowhere-vanishing $f \in C(S^1)$.

Similarly,

$$\ker(T_z^*) = \mathbb{C}\delta_0, \quad \operatorname{coker}(T_z^*) = 0,$$

so

$$\operatorname{Index}(T_z^*) = 1,$$

and the K -theory index class of T_z^* is $[p]$.

Proposition 6.2 (cf. [WY20, Prop. 2.7.15, Example 2.8.3, and Thm. 3.4.1]). *With the boundary convention in definition 5.5 and proposition 6.1,*

$$\partial_1([S^{-1}]) = [p], \quad \partial_1([S]) = -[p].$$

Hence

$$[S^{-1}] = \beta_1, \quad [S] = -\beta_1.$$

Proof. This choice fixes the sign: the positive half-line F is the I -term and the negative half-line E is the J -term. In the pushout description, take

$$A = C^*(\mathbb{R}), \quad I = C_X^*(F), \quad J = C_X^*(E).$$

Modulo the negative half-line ideal J , the stabilised representative

$$U_{S^{-1}} = 1 + (S^{-1} - 1) \otimes e$$

compresses on the positive half-line to

$$T_z^* \otimes e + 1 \otimes (1 - e).$$

The second summand is invertible and contributes zero to the index. The first summand has index projection $p \otimes e$, which is identified with p under the standard stabilisation $\mathcal{K} \otimes \mathcal{K} \cong \mathcal{K}$. The odd Mayer–Vietoris boundary is the index map for

$$0 \longrightarrow I \cap J \longrightarrow I \longrightarrow I/(I \cap J) \longrightarrow 0.$$

Therefore

$$\partial_1([S^{-1}]) = \text{Ind}[T_z^* \otimes e + 1 \otimes (1 - e)] = [p].$$

The same argument for S gives the compression $T_z \otimes e + 1 \otimes (1 - e)$, whose index class is $-[p]$. $\square$

7 Higher-dimensional shift products

7.1 The stabilised last-coordinate shift

Consider

$$\ell^2(\mathbb{Z}^n, H) \cong \ell^2(\mathbb{Z}^{n-1}) \otimes \ell^2(\mathbb{Z}) \otimes H \cong \ell^2(\mathbb{Z}^{n-1}, H) \otimes \ell^2(\mathbb{Z}),$$

where H is a separable infinite-dimensional Hilbert space, and the last factor records the last coordinate.

Let S_n be the bilateral shift in the last coordinate:

$$S_n \delta_{(m,k)} = \delta_{(m,k+1)}, \quad S_n^{-1} \delta_{(m,k)} = \delta_{(m,k-1)},$$

where $m \in \mathbb{Z}^{n-1}$ and $k \in \mathbb{Z}$.

By [WY20, Thm. 5.1.15], the inclusion $\mathbb{Z}^n \hookrightarrow \mathbb{R}^n$ is a coarse equivalence and induces an isomorphism

$$K_*(C^*(\mathbb{Z}^n)) \cong K_*(C^*(\mathbb{R}^n)).$$

So we may work with $\mathbb{Z}^n$.

The Roe algebra $C^*(\mathbb{Z}^n)$ is the norm closure of $C[H_{\mathbb{Z}^n}]$, where $H_{\mathbb{Z}^n} := \ell^2(\mathbb{Z}^n, H)$ and $C[H_{\mathbb{Z}^n}]$ consists of bounded finite-propagation matrices

$$(T_{xy})_{x,y \in \mathbb{Z}^n}, \quad T_{xy} \in \mathcal{K}(H).$$

The shift $S_n \otimes 1_H$ in the last coordinate has matrix entries

$$(S_n \otimes 1_H)_{(m,k),(m',k')} = \begin{cases} 1_H, & m = m', k = k' + 1, \\ 0, & \text{otherwise.} \end{cases}$$

Since 1_H is not compact, $S_n \otimes 1_H \notin C^*(\mathbb{Z}^n)$.

Let $e \in \mathcal{K}(H)$ be a rank-one projection. Then

$$((S_n^{-1} - 1) \otimes e)_{(m,k),(m',k')} = \begin{cases} -e, & (m,k) = (m',k'), \\ e, & m = m', k = k' - 1, \\ 0, & \text{otherwise.} \end{cases}$$

Now the matrix entries are compact and the propagation is finite. Hence

$$(S_n^{-1} - 1) \otimes e \in C^*(\mathbb{Z}^n),$$

and

$$U_{S_n^{-1}} := 1 + (S_n^{-1} - 1) \otimes e \in C^*(\mathbb{Z}^n)^+.$$

Equivalently,

$$U_{S_n^{-1}} = S_n^{-1} \otimes e + 1 \otimes (1 - e).$$

We have

$$U_{S_n^{-1}}^* = S_n \otimes e + 1 \otimes (1 - e).$$

Using

$$e(1 - e) = 0, \quad (1 - e)e = 0, \quad e^2 = e, \quad (1 - e)^2 = 1 - e,$$

we check that

$$U_{S_n^{-1}}^* U_{S_n^{-1}} = 1, \quad U_{S_n^{-1}} U_{S_n^{-1}}^* = 1.$$

Thus $U_{S_n^{-1}}$ is a unitary in $C^*(\mathbb{Z}^n)^+$, and it defines

$$[U_{S_n^{-1}}] \in K_1(C^*(\mathbb{Z}^n)).$$

7.2 Defining the higher shift class by products

For $n \geq 2$, the product representative for the recursively defined class is

$$\beta_n = (-1)^{n-1} m_*(\beta_{n-1} \times [S^{-1}]).$$

Here $[S^{-1}] \in K_1(C^*(\mathbb{Z}))$ is the one-dimensional shift class, placed in the last coordinate; the boundary computation below checks that this product representative has the recursive boundary value.

Here the product is the external product on spectral K -theory

$$\times : \text{sp } K_i(A) \otimes \text{sp } K_j(B) \longrightarrow \text{sp } K_{i+j}(A \widehat{\otimes}_{\max} B)$$

from [WY20, Def. 2.10.7]. We take $A = C^*(\mathbb{Z}^{n-1})$ and $B = C^*(\mathbb{Z})$. By [WY20, Prop. 2.9.12], spectral K -theory agrees with ordinary K -theory for the inner-graded algebras used here.

The product decomposition $\mathbb{Z}^n = \mathbb{Z}^{n-1} \times \mathbb{Z}$ gives commuting multiplier representations of the two Roe algebras on the product module. By the universal property of the maximal tensor product, these induce a

*-homomorphism

$$m : C^*(\mathbb{Z}^{n-1}) \widehat{\otimes}_{\max} C^*(\mathbb{Z}) \longrightarrow C^*(\mathbb{Z}^n).$$

On finite-propagation elementary tensors, this is the tensor-product operator on the product module, after the standard stabilisation

$$\mathcal{K}(H) \otimes \mathcal{K}(H) \cong \mathcal{K}(H).$$

Propagation remains finite, and compact matrix entries tensor to compact matrix entries. The separate factors $a \otimes 1$ and $1 \otimes b$ are generally multipliers rather than Roe operators, but elementary products $a \otimes b$ are Roe operators on the product module; this is the map that extends to m .

Thus the external product first gives a class in the tensor product, and m_* gives a class in the Roe algebra of $\mathbb{Z}^n$. The second factor $[S^{-1}]$ is the one-dimensional shift class; the resulting product also includes the first-factor class β_{n-1} .

If n is odd, then $\beta_{n-1} \in K_0(C^*(\mathbb{Z}^{n-1}))$. For a projection $q \in M_k(C^*(\mathbb{Z}^{n-1}))$ representing a K_0 -class, the product $[q] \times [S^{-1}]$ is represented by

$$U_n^q := q \otimes U_{S^{-1}} + (1 - q) \otimes 1,$$

where $U_{S^{-1}} = 1 + (S^{-1} - 1) \otimes e$ is the stabilised shift representative. Indeed,

$$(U_n^q)^* = q \otimes U_S + (1 - q) \otimes 1,$$

and

$$U_n^q (U_n^q)^* = (U_n^q)^* U_n^q = 1.$$

If

$$\chi = [q] - [q_0] \in K_0(C^*(\mathbb{Z}^{n-1})),$$

with q, q_0 in $M_k(A^+)$ and having the same image in $M_k(\mathbb{C})$, then $\chi \times [S^{-1}]$ is represented by

$$U_n^q (U_n^{q_0})^{-1}.$$

If n is even, then $\beta_{n-1} \in K_1(C^*(\mathbb{Z}^{n-1}))$. Let u be a unitary representative. The product

$$[u] \times [S^{-1}] \in K_0(C^*(\mathbb{Z}^{n-1}) \widehat{\otimes}_{\max} C^*(\mathbb{Z}))$$

is represented by a formal difference of projections using [WY20, Construction 2.9.11]. Here $C_0(\mathbb{R})$ has the grading $f(x) \mapsto f(-x)$, and $\mathcal{K}$ has the standard grading of [WY20, Example E.1.9]. Namely, let

$$\phi_u : C_0(\mathbb{R}) \longrightarrow C^*(\mathbb{Z}^{n-1}) \widehat{\otimes} \text{Cliff}_{\mathbb{C}}(\mathbb{R}) \widehat{\otimes} \mathcal{K}$$

and

$$\phi_{S^{-1}} : C_0(\mathbb{R}) \longrightarrow C^*(\mathbb{Z}) \widehat{\otimes} \text{Cliff}_{\mathbb{C}}(\mathbb{R}) \widehat{\otimes} \mathcal{K}$$

represent the two K_1 -classes. The external product is defined using the comultiplication

$$\Delta : C_0(\mathbb{R}) \longrightarrow C_0(\mathbb{R}) \widehat{\otimes} C_0(\mathbb{R}).$$

Then

$$\phi := (\phi_u \widehat{\otimes}_{\max} \phi_{S^{-1}}) \circ \Delta$$

with the canonical graded reorderings and stabilisation understood, represents a class in

$$\mathrm{sp} K_2(C^*(\mathbb{Z}^{n-1}) \widehat{\otimes}_{\max} C^*(\mathbb{Z})).$$

The graded isomorphism $\mathrm{Cliff}_{\mathbb{C}}(\mathbb{R}^2) \widehat{\otimes} \mathcal{K} \cong \mathcal{K}$ induces

$$\mathrm{sp} K_2(A \widehat{\otimes}_{\max} B) \cong \mathrm{sp} K_0(A \widehat{\otimes}_{\max} B)$$

by [WY20, Rem. 2.9.2]. Since A and B have trivial gradings, [WY20, Prop. 2.9.12] identifies the latter group with $K_0(A \widehat{\otimes}_{\max} B)$.

Using this graded isomorphism, apply [WY20, Construction 2.9.11] to view ϕ as a graded homomorphism into $M_2((C^*(\mathbb{Z}^{n-1}) \widehat{\otimes}_{\max} C^*(\mathbb{Z})) \otimes \mathcal{K})$. Here $\mathcal{K}$ denotes an ungraded copy of the compact operators, and the matrix grading is implemented by ν below. Let u_ϕ be its Cayley transform, and let

$$\nu = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

be the grading multiplier. Set

$$p_\phi := \frac{1}{2}(\nu u_\phi + 1).$$

Then

$$p_\phi \equiv \frac{1}{2}(\nu + 1) \pmod{M_2((C^*(\mathbb{Z}^{n-1}) \widehat{\otimes}_{\max} C^*(\mathbb{Z})) \otimes \mathcal{K})}.$$

Thus the corresponding ordinary K_0 -class is

$$\left[\frac{1}{2}(\nu + 1) \right] - [p_\phi] \in K_0(C^*(\mathbb{Z}^{n-1}) \widehat{\otimes}_{\max} C^*(\mathbb{Z})).$$

After applying m_* , this gives a class in $K_0(C^*(\mathbb{Z}^n))$. For example,

$$\beta_2 = -m_*([S^{-1}] \times [S^{-1}]) = m_*\left([p_\phi] - \left[\frac{1}{2}(\nu + 1)\right]\right) \in K_0(C^*(\mathbb{Z}^2)) \cong K_0(C^*(\mathbb{R}^2)).$$

7.3 Boundary computation for the odd-dimensional step

We give the concrete computation for the $K_0 \times K_1 \rightarrow K_1$ case. Work on

$$\ell^2(\mathbb{Z}^n, H) \cong \ell^2(\mathbb{Z}^{n-1}) \otimes \ell^2(\mathbb{Z}) \otimes H.$$

Let

$$\chi = [q] - [q_0] \in K_0(C^*(\mathbb{Z}^{n-1})),$$

where q, q_0 are projections after stabilising if necessary. In the non-unital case, they are understood in the unitisation, with matching scalar parts.

In that case, the individual index computations below take place in $K_0(M_k(C^*(\mathbb{Z}^{n-1})^+) \otimes \mathcal{K})$ for a common

matrix size k ; their difference has zero scalar image and thus belongs to $K_0(C^*(\mathbb{Z}^{n-1}) \otimes \mathcal{K})$.

Define

$$U_n^q := q \otimes U_{S^{-1}} + (1 - q) \otimes 1, \quad U_n^{q_0} := q_0 \otimes U_{S^{-1}} + (1 - q_0) \otimes 1.$$

Then the class $\chi \times [S^{-1}]$ is represented by

$$U_n^q (U_n^{q_0})^{-1}.$$

Let

$$V : \ell^2(\mathbb{N}) \hookrightarrow \ell^2(\mathbb{Z}), \quad V\delta_k = \delta_k,$$

and put

$$P_+ := VV^*, \quad P_n^+ := 1 \otimes P_+ \otimes 1_H.$$

Identifying the range of P_n^+ with

$$\ell^2(\mathbb{Z}^{n-1}) \otimes \ell^2(\mathbb{N}) \otimes H,$$

the positive-half-space compression of U_n^q is

$$\widehat{U} = q \otimes (T_z^* \otimes e + 1 \otimes (1 - e)) + (1 - q) \otimes 1.$$

The summand with $1 - e$ is invertible and has zero index contribution. Removing the identity summand does not change the index. After stabilisation, it suffices to compute the index of

$$\widehat{U} = q \otimes T_z^* + (1 - q) \otimes 1.$$

Take

$$W := q \otimes T_z + (1 - q) \otimes 1$$

as a parametrix for this reduced operator. Since $T_z^* T_z = 1$ and $T_z T_z^* = 1 - p$, where p is the rank-one projection onto $\mathbb{C}\delta_0$, we get

$$\widehat{U}W = 1, \quad W\widehat{U} = 1 - q \otimes p.$$

Thus

$$1 - \widehat{U}W = 0, \quad 1 - W\widehat{U} = q \otimes p.$$

Using the algebraic index formula from [WY20, Rem. 2.8.2, Eq. (2.20)],

$$\text{Ind}[\widehat{U}] = \left[\begin{pmatrix} (1 - W\widehat{U})^2 & W(1 - \widehat{U}W) \\ \widehat{U}(2 - W\widehat{U})(1 - W\widehat{U}) & \widehat{U}W(2 - \widehat{U}W) \end{pmatrix} \right] - \left[\begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix} \right].$$

Substituting

$$1 - W\widehat{U} = q \otimes p, \quad 1 - \widehat{U}W = 0, \quad \widehat{U}W = 1,$$

we obtain

$$(1 - W\widehat{U})^2 = q \otimes p, \quad W(1 - \widehat{U}W) = 0,$$

and

$$\widehat{U}W(2 - \widehat{U}W) = 1.$$

For the lower-left entry,

$$(2 - W\widehat{U})(1 - W\widehat{U}) = (1 + q \otimes p)(q \otimes p) = 2q \otimes p.$$

Hence

$$\widehat{U}(2 - W\widehat{U})(1 - W\widehat{U}) = 2\widehat{U}(q \otimes p) = 2q \otimes T_z^*p = 0,$$

because $T_z^*\delta_0 = 0$. Therefore

$$\text{Ind}[\widehat{U}] = \left[\begin{pmatrix} q \otimes p & 0 \\ 0 & 1 \end{pmatrix} \right] - \left[\begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix} \right] = [q \otimes p].$$

Similarly,

$$\text{Ind}[\widehat{U}^{q_0}] = [q_0 \otimes p].$$

Let $W^{q_0} := q_0 \otimes T_z + (1 - q_0) \otimes 1$ be the corresponding parametrix. Then $\widehat{U}W^{q_0}$ is the compression of $U_n^q(U_n^{q_0})^{-1}$, after removing the identity summand. By additivity of the index,

$$\text{Ind}[\widehat{U}W^{q_0}] = [q \otimes p] - [q_0 \otimes p].$$

By stabilisation, this corresponds to $[q] - [q_0] = \chi$. Therefore

$$\partial_n(m_*(\chi \times [S^{-1}])) = \chi.$$

Taking $\chi = \beta_{n-1}$ gives

$$\partial_n(\beta_n) = \beta_{n-1}, \quad \beta_n = m_*(\beta_{n-1} \times [S^{-1}])$$

in the odd-dimensional step where $\beta_{n-1} \in K_0$.

For the forward shift S , the compression is

$$\widehat{U}_{\text{forw}} = q \otimes T_z + (1 - q) \otimes 1,$$

and a parametrix is

$$W_{\text{forw}} = q \otimes T_z^* + (1 - q) \otimes 1.$$

Then

$$W_{\text{forw}}\widehat{U}_{\text{forw}} = 1, \quad \widehat{U}_{\text{forw}}W_{\text{forw}} = 1 - q \otimes p.$$

Thus

$$1 - W_{\text{forw}}\widehat{U}_{\text{forw}} = 0, \quad 1 - \widehat{U}_{\text{forw}}W_{\text{forw}} = q \otimes p.$$

Using the same algebraic index formula,

$$(1 - W_{\text{forw}}\widehat{U}_{\text{forw}})^2 = 0,$$

$$W_{\text{forw}}(1 - \widehat{U}_{\text{forw}}W_{\text{forw}}) = W_{\text{forw}}(q \otimes p) = q \otimes T_z^*p = 0,$$

$$\widehat{U}_{\text{forw}}(2 - W_{\text{forw}}\widehat{U}_{\text{forw}})(1 - W_{\text{forw}}\widehat{U}_{\text{forw}}) = 0,$$

and

$$\widehat{U}_{\text{forw}} W_{\text{forw}} (2 - \widehat{U}_{\text{forw}} W_{\text{forw}}) = (1 - q \otimes p)(1 + q \otimes p) = 1 - q \otimes p.$$

Therefore the index projection becomes

$$\left[\begin{pmatrix} 0 & 0 \\ 0 & 1 - q \otimes p \end{pmatrix} \right] - \left[\begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix} \right] = -[q \otimes p].$$

Therefore

$$\partial_n(m_*(\chi \times [S])) = -\chi.$$

So the backward shift gives the positive boundary value, and the forward shift gives the negative boundary value.

For even n , the first factor has degree one. Using the index and exponential maps of [WY20, Rem. 2.6.8 and Def. 2.8.4] and the external product of [WY20, Def. 2.10.7] gives the boundary formula

$$\partial_n(m_*(\chi \times [S^{-1}])) = (-1)^{n-1} \chi, \quad \chi \in K_{n-1}(C^*(\mathbb{Z}^{n-1})).$$

Thus the formula verified above has a positive sign when n is odd and a negative sign when n is even. To retain the recursive convention $\partial_n(\beta_n) = \beta_{n-1}$, we therefore have

$$\beta_n = (-1)^{n-1} m_*(\beta_{n-1} \times [S^{-1}]).$$

8 The Dirac picture and comparison with the shift generator

8.1 The Euclidean Dirac class

Definition 8.1 ([WY20, Def. 9.3.1]). Let $n \in \mathbb{N}$, and let H_n be the Hilbert space underlying the Clifford algebra $\text{Cliff}_{\mathbb{C}}(\mathbb{R}^n)$. Fix an orthonormal basis $\{e_1, \dots, e_n\}$ for $\mathbb{R}^n$, and let $x_1, \dots, x_n$ be the corresponding coordinates.

For $v \in \mathbb{R}^n$, write $\widehat{v}$ for the operator on $\text{Cliff}_{\mathbb{C}}(\mathbb{R}^n)$ defined on a homogeneous Clifford element w by

$$\widehat{v}(w) = (-1)^{\deg(w)} wv.$$

Thus $\widehat{v}$ is right multiplication by v , twisted by the grading. The Dirac operator D_n on $L^2(\mathbb{R}^n; H_n)$ has domain $\mathcal{S}(\mathbb{R}^n; H_n)$ and is given by

$$(D_n u)(x) = \sum_{i=1}^n \widehat{e}_i \frac{\partial u}{\partial x_i}(x).$$

The Bott operator C_n on the same domain is

$$(C_n u)(x) = \sum_{i=1}^n e_i x_i u(x).$$

Let $\psi : \mathbb{R} \rightarrow [-1, 1]$ be a continuous odd function with

$$\lim_{x \rightarrow \pm\infty} \psi(x) = \pm 1.$$

Set $\mathcal{F}_n := \psi(D_n)$. By the functional calculus, $\mathcal{F}_n$ is a self-adjoint multiplier and $1 - \mathcal{F}_n^2 \in C^*(\mathbb{R}^n)$ [WY20, Lem. 3.2.7]. We use the graded $*$ -homomorphism defined by the functional calculus of D_n , with the Clifford factor as in [WY20, Sec. 9.3], and the isomorphism with ordinary K -theory from [WY20, Prop. 2.9.12]. In even dimensions,

$$\text{Index}[\mathcal{F}_{2m}] \in K_0(C^*(\mathbb{R}^{2m})),$$

and in odd dimensions,

$$\text{Index}[\mathcal{F}_{2m+1}] \in K_1(C^*(\mathbb{R}^{2m+1})).$$

Let

$$[D_n] \in K_n(L^*(\mathbb{R}^n)) = K_n(\mathbb{R}^n)$$

be the K -homology Dirac class, and let

$$[C_n] \in K^n(\mathbb{R}^n)$$

be the Bott class. By [WY20, Lem. 9.3.4],

$$[D_n] = [D_1] \times \cdots \times [D_1], \quad [C_n] = [C_1] \times \cdots \times [C_1],$$

where the products are exterior products. By [WY20, Thm. 9.3.5],

$$\langle [D_n], [C_n] \rangle = 1 \in \mathbb{Z}.$$

If $\alpha \in K_n(\mathbb{R}^n)$ and $\gamma \in K^n(\mathbb{R}^n)$ are generators, and if

$$[D_n] = m\alpha, \quad [C_n] = r\gamma,$$

then

$$1 = \langle [D_n], [C_n] \rangle = mr\langle \alpha, \gamma \rangle.$$

Hence $|m| = |r| = 1$, and $[D_n]$ is a generator of $K_n(\mathbb{R}^n)$.

By corollary 5.4, the evaluation-at-one map

$$\text{ev} : C_L^*(\mathbb{R}^n) \longrightarrow C^*(\mathbb{R}^n)$$

induces an isomorphism on K -theory. Also, by proposition 1.8, the localised Roe algebra computes K -homology for ample modules. Therefore the Dirac class gives a localised index class represented by the family

$$\mathcal{F}_{n,t}^L := \psi(t^{-1}D_n), \quad t \in [1, \infty),$$

namely

$$\text{Index}_L[(\mathcal{F}_{n,t}^L)] \in K_n(C_L^*(\mathbb{R}^n)),$$

and evaluation at $t = 1$ gives

$$\text{ev}_*(\text{Index}_L[(\mathcal{F}_{n,t}^L)]) = \text{Index}[\mathcal{F}_n] \in K_n(C^*(\mathbb{R}^n)).$$

Since $[D_n]$ is a generator and ev_* is an isomorphism, $\text{Index}[\mathcal{F}_n]$ is a generator. Consequently,

$$\beta_n = \pm \text{Index}[\mathcal{F}_n] = \pm \text{Index}[\psi(D_n)].$$

The generator argument determines this equality only up to sign; the exact sign requires a comparison with the fixed Mayer–Vietoris convention.

Under the decomposition $\mathbb{R}^n = \mathbb{R}^{n-1} \times \mathbb{R}$, [WY20, Rem. 9.3.2] gives

$$D_n = D_{n-1} \widehat{\otimes} 1 + 1 \widehat{\otimes} D_1.$$

The product decomposition alone does not fix the sign of the Dirac class under the boundary map. The shift generator retains the convention

$$\partial_n(\beta_n) = \beta_{n-1}.$$

8.2 Comparison with the shift product

The dimension-one calculation fixes the shift sign:

$$\partial_1([S^{-1}]) = [p].$$

Both $[S^{-1}]$ and $\text{Index}[\psi(D_1)]$ generate $K_1(C^*(\mathbb{R}))$, so they agree up to sign. Iterating the product formula for the recursively defined shift generator gives

$$\beta_n = (-1)^{n(n-1)/2} m_* \left(\underbrace{[S^{-1}] \times \cdots \times [S^{-1}]}_{n \text{ factors}} \right).$$

Here each factor is the one-dimensional shift class in its corresponding coordinate. The Dirac product formula uses $D_n = D_{n-1} \widehat{\otimes} 1 + 1 \widehat{\otimes} D_1$. Thus the shift product and the Dirac class both generate the same group; an exact signed identification between them is not established by the calculations above. The shift generator satisfies

$$\partial_n(\beta_n) = \beta_{n-1}.$$

In particular, it is obtained from the external product with the backward shift class, with the parity sign specified above.

8.3 Why the Dirac operator looks like a shift

The harmonic oscillator is

$$H = -\frac{d^2}{dx^2} + x^2 - 1$$

on $L^2(\mathbb{R})$, with domain the Schwartz-class functions $\mathcal{S}(\mathbb{R})$ [WY20, Def. D.3.1]. The annihilation and creation operators are

$$A = x + \frac{d}{dx}, \quad A^* = x - \frac{d}{dx},$$

considered on the domain $\mathcal{S}(\mathbb{R})$, and they satisfy

$$H = A^* A = A A^* - 2.$$

Define

$$\psi_0(x) = \pi^{-1/4} e^{-x^2/2}, \quad \psi_k = \frac{1}{\sqrt{2^k k!}} (A^*)^k \psi_0.$$

By [WY20, Prop. D.3.3], $(\psi_k)_{k=0}^\infty$ is an orthonormal basis of $L^2(\mathbb{R})$ consisting of eigenvectors for H , with

$$H\psi_k = 2k\psi_k.$$

We compute $\psi'_0(x) = -x\psi_0(x)$, so

$$A\psi_0 = \left(\frac{d}{dx} + x\right)\psi_0 = \psi'_0 + x\psi_0 = 0.$$

For $n \geq 1$,

$$A\psi_n = \sqrt{2n}\psi_{n-1}.$$

Thus

$$A\psi_0 = 0, \quad A\psi_1 = \sqrt{2}\psi_0, \quad A\psi_2 = 2\psi_1, \quad A\psi_3 = \sqrt{6}\psi_2, \dots$$

Compare this with the backward unilateral shift T_z^* :

$$T_z^* \delta_0 = 0, \quad T_z^* \delta_1 = \delta_0, \quad T_z^* \delta_2 = \delta_1, \dots$$

Thus, in the Hermite basis, A is a weighted backward shift. The one-dimensional Bott–Dirac operator appearing in the proof of Bott periodicity is

$$B = \begin{pmatrix} 0 & A^* \\ A & 0 \end{pmatrix} = \begin{pmatrix} 0 & -\frac{d}{dx} + x \\ \frac{d}{dx} + x & 0 \end{pmatrix},$$

as in [WY20, proof of Thm. 9.3.5]. Since $A\psi_0 = 0$ and $A\psi_n = \sqrt{2n}\psi_{n-1}$ for $n \geq 1$,

$$\ker(A) = \mathbb{C}\psi_0.$$

Moreover, every basis vector ψ_k lies in the range of A because

$$A \left(\frac{1}{\sqrt{2(k+1)}} \psi_{k+1} \right) = \psi_k.$$

For the closure of A on $L^2(\mathbb{R})$, the assignment

$$\sum_{k \geq 0} a_k \psi_k \mapsto \sum_{k \geq 0} \frac{a_k}{\sqrt{2(k+1)}} \psi_{k+1}$$

is a bounded right inverse: the image lies in the domain of the closure because

$$\sum_{k \geq 0} 2(k+1) \left| \frac{a_k}{\sqrt{2(k+1)}} \right|^2 = \sum_{k \geq 0} |a_k|^2 < \infty.$$

Thus the closed operator is surjective with one-dimensional kernel, and

$$\text{Index}(A) = \dim \ker(A) - \dim \text{coker}(A) = 1.$$

The backward unilateral shift has the same index:

$$\text{Index}(T_z^*) = 1.$$

So we have the correspondences

$$T_z^* \longleftrightarrow A = \frac{d}{dx} + x, \quad \delta_0 \longleftrightarrow \psi_0(x) = \pi^{-1/4} e^{-x^2/2}, \quad [p_{\delta_0}] \longleftrightarrow [p_{\psi_0}].$$

Thus both T_z^* and A have index 1.

9 Assembly, Rips complexes, and coarse K -homology

This section records the background, using the definitions of Roe and localised Roe algebras from definitions 1.4 and 1.7.

9.1 Assembly maps and coarse K -homology

Let X be a proper metric space, let H_X be an ample X module, and assume first that the group action is trivial. The evaluation-at-one $*$ -homomorphism

$$\text{ev} : C_L^*(H_X) \longrightarrow C^*(H_X), \quad (T_t) \longmapsto T_1$$

induces, after the identification of proposition 1.8, the assembly map.

Definition 9.1 ([WY20, Def. 7.1.1]). The *assembly map*, or *higher index map*, for X is

$$\mu_X : K_*(X) \longrightarrow K_*(C^*(X)).$$

It is induced by the evaluation-at-one map on localised Roe algebras.

In the equivariant setting, X is equipped with a proper isometric action of a countable discrete group G , and the same definitions use equivariant modules and equivariant Roe algebras. We then write

$$\mu_X : K_*^G(X) \longrightarrow K_*(C^*(X)^G).$$

Definition 9.2 ([WY20, Def. 7.1.4]). The equivariant bounded category over X , denoted $C^G(X)$, has objects pairs (Y, p) , where Y is a proper metric space with a proper isometric G -action and $p : X \rightarrow Y$ is an equivariant coarse equivalence. A morphism from (Y, p_Y) to (Z, p_Z) is a continuous equivariant coarse map $f : Y \rightarrow Z$ such that $f \circ p_Y$ is close to p_Z .

Definition 9.3 ([WY20, Defs. 7.1.7, 7.1.8, 7.1.10, and 7.1.11]). The equivariant coarse K -homology of X , denoted $KX_*^G(X)$, is the universal F -group for the functor $F(Y) := K_*^G(Y)$ on $C^G(X)$. For an object (Y, p)

in $C^G(X)$, the X -assembly map is

$$\mu_{Y,X} : K_*^G(Y) \xrightarrow{\mu_Y} K_*(C^*(Y)^G) \xrightarrow{(p_*)^{-1}} K_*(C^*(X)^G).$$

The Baum–Connes assembly map is the homomorphism

$$\mu : KX_*^G(X) \longrightarrow K_*(C^*(X)^G)$$

induced by the universal property of $KX_*^G(X)$. The Baum–Connes conjecture for X says that this map is an isomorphism. If $G = \{e\}$, we omit G and call

$$\mu : KX_*(X) \longrightarrow K_*(C^*(X))$$

the coarse Baum–Connes assembly map.

9.2 Rips complexes and the direct-limit model

Definition 9.4 ([WY20, Def. A.3.10]). Let $r \in (0, \infty)$. A subset $Z \subseteq X$ is r -separated if

$$d(x, y) \geq r \quad \text{for all distinct } x, y \in Z.$$

It is an r -net if it is r -separated and, for every $x \in X$, there is $z \in Z$ with $d(x, z) < r$. By [WY20, Lem. A.3.11], every proper metric space admits an r -net for any $r \in (0, \infty)$. Moreover, the restricted metric on any net is proper, and the inclusion $Z \hookrightarrow X$ is a coarse equivalence.

Definition 9.5 ([WY20, Defs. 7.2.1, 7.2.2, and 7.2.8]). If F is a finite set, let $\sigma(F)$ be the set of formal sums

$$\sum_{z \in F} t_z z, \quad t_z \in [0, 1], \quad \sum_{z \in F} t_z = 1.$$

Let $S(\mathbb{R}^F)$ be the unit sphere in the finite-dimensional Euclidean space $\mathbb{R}^F$ spanned by F . Embed $\sigma(F)$ into $S(\mathbb{R}^F)$ by

$$\sum_{z \in F} t_z z \mapsto \left(\sum_{z \in F} t_z^2 \right)^{-1/2} \sum_{z \in F} t_z z.$$

The spherical metric on $\sigma(F)$ is

$$d(x, y) := \frac{2}{\pi} d_{\text{in}}(x, y),$$

where d_{in} is the intrinsic metric on the sphere after the above embedding. The factor $2/\pi$ normalises the diameter of each positive-dimensional spherical simplex to be 1.

Let Z be a locally finite metric space and let $r \geq 0$. The spherical Rips complex $S_r(Z)$ consists of all formal sums

$$x = \sum_{z \in Z} t_z z$$

such that $t_z \in [0, 1]$, $\sum_z t_z = 1$, and $\text{diam}(\text{supp}(x)) \leq r$, where $\text{supp}(x) := \{z \in Z : t_z \neq 0\}$. If $F \subseteq Z$ is finite and $\text{diam}(F) \leq r$, the simplex spanned by F is the subset $\sigma(F) \subseteq S_r(Z)$ of formal sums supported in F , with the spherical metric.

A simplicial path γ from x to y in $S_r(Z)$ is a finite sequence $x = x_0, x_1, \dots, x_n = y$, together with simplices $\sigma_1, \dots, \sigma_n$ such that σ_i contains x_{i-1} and x_i . Its length is

$$\ell(\gamma) := \sum_{i=1}^n d_{\sigma_i}(x_{i-1}, x_i),$$

and the spherical distance is

$$d_{S_r}(x, y) := \inf\{\ell(\gamma) : \gamma \text{ is a simplicial path from } x \text{ to } y\},$$

with $d_{S_r}(x, y) = \infty$ if no simplicial path exists.

A semi-simplicial path δ from x to y in $S_r(Z)$ is a sequence

$$x = x_0, y_0, x_1, y_1, \dots, x_n, y_n = y,$$

where $x_1, \dots, x_n \in Z$ and $y_0, \dots, y_{n-1} \in Z$. Its length is

$$\ell(\delta) := \sum_{i=0}^n d_{S_r}(x_i, y_i) + \sum_{i=0}^{n-1} d_Z(y_i, x_{i+1}).$$

The semi-spherical distance is

$$d_{P_r}(x, y) := \inf\{\ell(\delta) : \delta \text{ is a semi-simplicial path from } x \text{ to } y\}.$$

The Rips complex $P_r(Z)$ is the set $S_r(Z)$ equipped with the metric d_{P_r} . In particular, $P_0(Z) = Z$.

Proposition 9.6 ([WY20, Prop. 7.2.11]). *If Z is countable and locally finite, then $P_r(Z)$ is a proper, second countable metric space. For every $s \geq r$, the canonical inclusion*

$$i_{rs} : P_r(Z) \longrightarrow P_s(Z)$$

is a homeomorphism onto its image and a coarse equivalence.

Theorem 9.7 ([WY20, Thm. 7.2.16]). *Let X be a proper metric space with a proper isometric action of a countable discrete group G , and let $Z \subseteq X$ be a locally finite G -invariant net. Then*

$$KX_*^G(X) \cong \lim_{r \rightarrow \infty} K_*^G(P_r(Z)).$$

Under this identification, the Baum–Connes assembly map is the direct limit of the X -assembly maps

$$\lim_{r \rightarrow \infty} \mu_{P_r(Z), X} : \lim_{r \rightarrow \infty} K_*^G(P_r(Z)) \longrightarrow K_*(C^*(X))^G.$$

If $G = \{e\}$, then

$$KX_*(X) \cong \lim_{r \rightarrow \infty} K_*(P_r(Z))$$

and

$$\mu = \lim_{r \rightarrow \infty} \mu_{P_r(Z), X} : \lim_{r \rightarrow \infty} K_*(P_r(Z)) \longrightarrow K_*(C^*(X)).$$

9.3 Generators from Rips complexes

When the coarse assembly map for X is an isomorphism, every class $\alpha \in K_i(C^*(X))$ has the form

$$\alpha = \mu_{P_r(Z), X}(\alpha_r)$$

for some scale $r \geq 0$ and some $\alpha_r \in K_i(P_r(Z))$. Under this identification, representatives

$$\alpha_r \in K_i(P_r(Z)), \quad \beta_s \in K_i(P_s(Z))$$

represent the same element of $K_i(C^*(X))$ if and only if there is $t \geq \max\{r, s\}$ such that

$$(i_{rt})_*(\alpha_r) = (i_{st})_*(\beta_s) \quad \text{in } K_i(P_t(Z)).$$

This is exactly the equivalence relation in the direct limit. Thus $K_i(P_r(Z))$ may have several independent classes at a fixed scale, but the maps

$$(i_{rs})_* : K_i(P_r(Z)) \longrightarrow K_i(P_s(Z))$$

may identify or kill some of these classes. The resulting direct limit may be noncyclic.

9.4 Bounded geometry and coarse assembly

Definition 9.8 ([WY20, Def. A.3.19]). A metric space X has bounded geometry if, for every $r \in (0, \infty)$, there is $n_r \in \mathbb{N}$ such that

$$|B(x; r)| \leq n_r \quad \text{for all } x \in X.$$

In particular, a bounded geometry metric space is proper and discrete.

Theorem 9.9 ([WY20, Thm. 12.0.2]). *If X is a bounded geometry metric space that coarsely embeds into a Hilbert space, then the coarse assembly map*

$$\mu : KX_*(X) \xrightarrow{\cong} K_*(C^*(X))$$

is an isomorphism. Therefore, for $i = 0, 1$,

$$K_i(C^*(X)) \cong KX_i(X) \cong \lim_{r \rightarrow \infty} K_i(P_r(Z)),$$

where $Z \subseteq X$ is any locally finite net.

Definition 9.10 ([WY20, Defs. 7.3.1 and 7.3.4]). A proper metric space X is *uniformly contractible* if, for every $r \geq 0$, there is $s \geq r$ such that, for every $x \in X$, the inclusion

$$B(x; r) \hookrightarrow B(x; s)$$

is homotopic to a constant map. A simplicial complex structure on X is *good* if it is finite-dimensional, if the vertex set has bounded geometry, and if the inclusion of the vertex set into X is a coarse equivalence.

Theorem 9.11 ([WY20, Thm. 7.3.6]). *If X is a uniformly contractible good simplicial complex, then the*

coarse Baum–Connes conjecture for X is equivalent to the assertion that the assembly map for X itself

$$\mu_X : K_*(X) \longrightarrow K_*(C^*(X))$$

is an isomorphism. So if X is uniformly contractible and good, and if the coarse assembly map is known to be an isomorphism, then

$$K_i(C^*(X)) \cong K_i(X), \quad i = 0, 1.$$

9.5 Elliptic operators and K -homology classes

We now recall how elliptic differential operators give analytic K -homology classes. Here M is a smooth Riemannian manifold, $S \rightarrow M$ is a smooth finite-rank Hermitian bundle, and differential operators are first-order and linear, as in [WY20, Def. 8.1.1].

Definition 9.12 ([WY20, Def. 8.3.1]). Let D be a differential operator on a Hermitian bundle $S \rightarrow M$. The operator D is elliptic if its symbol $\sigma_D(x, \xi) : S_x \rightarrow S_x$ is invertible for every nonzero cotangent vector $\xi \in T_x^*M$.

A differentiable function $f : \mathbb{R} \rightarrow \mathbb{C}$ has slow oscillation at infinity if $\sup_{x \in \mathbb{R}} |xf'(x)| < \infty$. The algebra $C_{\text{so}}(\mathbb{R})$ is the C^* -subalgebra of $C_b(\mathbb{R})$ generated by bounded functions with slow oscillation at infinity [WY20, Def. 8.2.4].

For a formally self-adjoint D , [WY20, Lem. 8.2.8] gives a family $(g_t)_{t \in [1, \infty)}$ of smooth functions $g_t : M \rightarrow [0, 1]$ such that:

- (i) the map $t \mapsto g_t$ is norm continuous as a function $[1, \infty) \rightarrow C_b(M)$;
- (ii) for every compact $K \subseteq M$, there is t_K such that $g_t = 1$ on K for all $t \geq t_K$;
- (iii) each operator $g_t D g_t$, with domain $C_c^\infty(M; S)$, is essentially self-adjoint.

If $f \in C_{\text{so}}(\mathbb{R})$, then the pair $(f, (g_t))$ is multiplier data for D in the sense of [WY20, Def. 8.2.9].

Theorem 9.13 ([WY20, Thm. 8.2.10 and Thm. 8.3.3]). *Let D be a formally self-adjoint differential operator, and let $(f, (g_t))$ be multiplier data for D . Define*

$$F_t := f(t^{-1} g_t D g_t).$$

Then $(F_t)_{t \in [1, \infty)}$ defines a multiplier of $L^(L^2(M; S))$. If D is elliptic and, in addition, $f \in C_0(\mathbb{R})$, then*

$$(F_t)_{t \in [1, \infty)} \in L^*(L^2(M; S)).$$

Proposition 9.14 ([WY20, Thm. 8.2.10, Thm. 8.3.3, Constructions 8.3.11 and 8.3.12, Prop. 8.3.13]). *Let D be a formally self-adjoint elliptic differential operator as in definition 9.12, and choose a family (g_t) as above. In the odd case, the map*

$$\Phi : C_0(\mathbb{R}) \longrightarrow L^*(L^2(M; S)), \quad h \longmapsto (h(t^{-1} g_t D g_t))_{t \in [1, \infty)}$$

is a $$ -homomorphism by [WY20, Construction 8.3.11], using theorem 9.13 together with functional calculus.*

Tensoring with a rank-one projection $p \in \mathcal{K}$ gives

$$\Phi \otimes p : C_0(\mathbb{R}) \longrightarrow L^*(L^2(M; S)) \otimes \mathcal{K},$$

which defines

$$[D] \in K_1(L^*(L^2(M; S))).$$

If $L^2(M; S)$ is an ample M module, then this is a class $[D] \in K_1(M)$.

In the even case, assume $S = S^- \oplus S^+$ is $\mathbb{Z}/2$ -graded, and that D interchanges $C_c^\infty(M; S^-)$ and $C_c^\infty(M; S^+)$. Let U be the grading operator on

$$L^2(M; S) = L^2(M; S^-) \oplus L^2(M; S^+),$$

acting by multiplication by -1 on $L^2(M; S^-)$ and by $+1$ on $L^2(M; S^+)$. Conjugation by U gives an inner grading on $L^*(L^2(M; S))$, and D is odd for this grading. The algebra $C_0(\mathbb{R})$ is graded by even and odd functions. Tensoring the graded $*$ -homomorphism above with a rank-one projection in the standard graded compact operators whose range lies in the even subspace gives a class

$$[D] \in K_0(L^*(L^2(M; S))).$$

If $L^2(M; S)$ is an ample M module, this identifies with a class in $K_0(M)$. The resulting classes $[D] \in K_i(M)$ do not depend on the choices of the family (g_i) or the rank-one projection.

9.6 Boundary maps at the level of coarse K -homology

Let $X = E \cup F$ be a coarsely excisive decomposition by closed subsets. Exercise 7.6.14 of [WY20] gives a commutative diagram of Mayer–Vietoris sequences:

$$\begin{array}{ccccccc} KX_i(E \cap F) & \longrightarrow & KX_i(E) \oplus KX_i(F) & \longrightarrow & KX_i(X) & \xrightarrow{\partial_{KX}} & KX_{i+1}(E \cap F) \\ \mu_{E \cap F} \downarrow & & \downarrow \mu_E \oplus \mu_F & & \downarrow \mu_X & & \downarrow \mu_{E \cap F} \\ K_i(C^*(E \cap F)) & \rightarrow & K_i(C^*(E)) \oplus K_i(C^*(F)) & \rightarrow & K_i(C^*(X)) & \xrightarrow{\partial_{Rqe}} & K_{i+1}(C^*(E \cap F)). \end{array}$$

The vertical maps are the coarse Baum–Connes assembly maps.

Let $Z \subseteq X$ be a net and put

$$Z_E := Z \cap E, \quad Z_F := Z \cap F.$$

In general,

$$P_r(Z) \neq P_r(Z_E) \cup P_r(Z_F).$$

Indeed, a simplex in $P_r(Z)$ may have all vertices of mutual Z -diameter at most r , with some vertices lying in $E \setminus F$ and others lying in $F \setminus E$.

So we thicken the vertex sets. For $R > r$, define

$$Z_E^{(R)} := Z \cap N_R(E), \quad Z_F^{(R)} := Z \cap N_R(F),$$

where $N_R(\cdot)$ denotes the R -neighbourhood. Then

$$P_r(Z) = P_r(Z_E^{(R)}) \cup P_r(Z_F^{(R)}).$$

To see this, let

$$x = \sum_{z \in Z} t_z z \in P_r(Z).$$

Choose a vertex $z_0 \in \text{supp}(x)$. Since $X = E \cup F$, either $z_0 \in E$ or $z_0 \in F$. If $z_0 \in E$, then every vertex $z \in \text{supp}(x)$ satisfies

$$d(z, E) \leq d(z, z_0) \leq r < R.$$

Thus the whole support of x lies in $Z_E^{(R)}$, and so $x \in P_r(Z_E^{(R)})$. The case $z_0 \in F$ is the same.

The vertex intersection is

$$Z_E^{(R)} \cap Z_F^{(R)} = Z \cap N_R(E) \cap N_R(F).$$

By coarse excisiveness, for this R there is $S > 0$ such that

$$N_R(E) \cap N_R(F) \subseteq N_S(E \cap F).$$

Therefore

$$Z_E^{(R)} \cap Z_F^{(R)} \subseteq Z \cap N_S(E \cap F).$$

Bounded neighbourhoods of $E \cap F$ are coarsely equivalent to $E \cap F$. After passing to direct limits as $r, R \rightarrow \infty$ with $R > r$, the term for the intersection is the coarse K -homology group $KX_*(E \cap F)$. Thus the commutative Mayer–Vietoris diagram gives, for every $b \in KX_i(X)$, the identity

$$\partial_{\text{Roe}}(\mu_X(b)) = \mu_{E \cap F}(\partial_{\text{KX}}(b)),$$

where

$$\partial_{\text{KX}} : KX_i(X) \longrightarrow KX_{i+1}(E \cap F)$$

is the Mayer–Vietoris boundary map. Equivalently, if μ_X is an isomorphism and $a \in K_i(C^*(X))$, then

$$\partial_{\text{Roe}}(a) = \mu_{E \cap F}(\partial_{\text{KX}}(\mu_X^{-1}(a))).$$

References

[WY20] Rufus Willett and Guoliang Yu, *Higher Index Theory*, Cambridge Studies in Advanced Mathematics, vol. 189, Cambridge University Press, 2020.